

Field of Dental Medicine

Biomaterials II

Lecture : (Casting Technology)

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Casting technology

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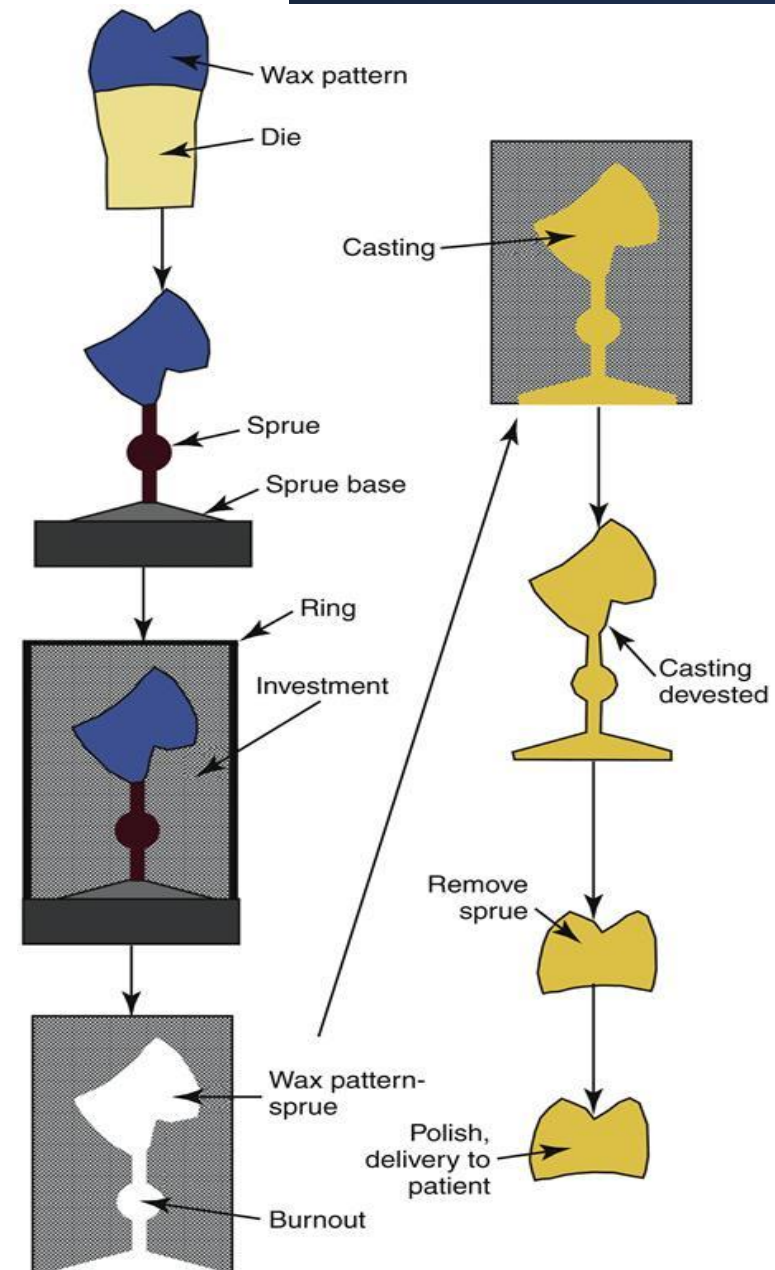
introduction

In dentistry, casting is a process by which a shaped wax pattern of a dental restoration is converted into alloy or ceramic restoration

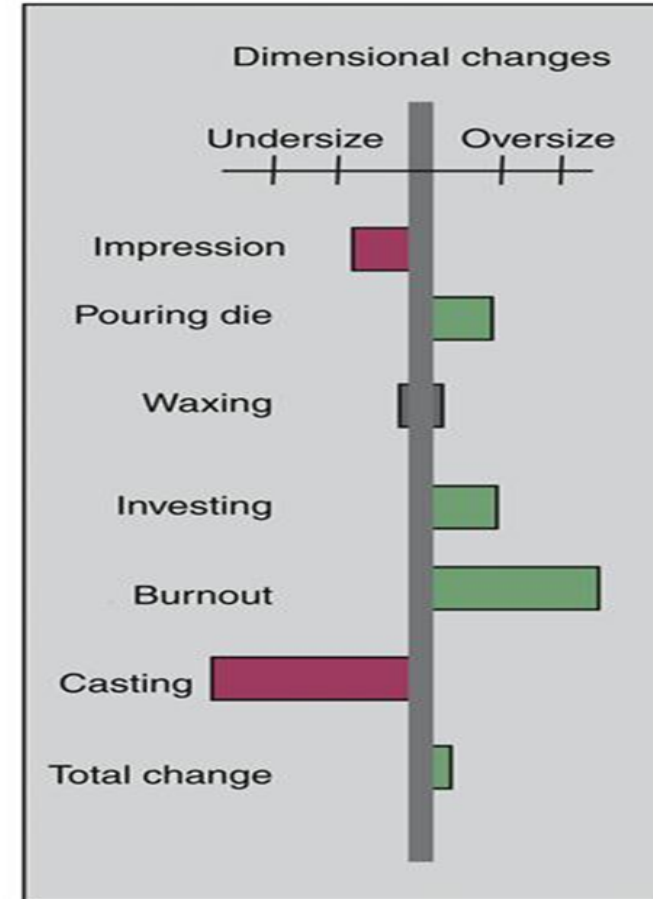
casting often uses a technique called the lost-wax technique. Figure 12-1 shows summary of the steps involved in the lost-wax technique: making the wax pattern, spruing the pattern, investing, burnout, casting, removal of investment (devesting), and finishing.

• **FIG 12-1 Sequence of steps for casting using the lost-wax technique:**

1. The wax pattern is formed on the die.
2. A sprue (generally made of wax or plastic) is added and the wax pattern is transferred to a sprue base.
3. A ring is added and the ring is filled with investment, surrounding the wax pattern and sprue.
4. The sprue base is removed, and the ring is placed into a hot oven that burns away the wax pattern and sprue.
5. Molten alloy is cast into the space formerly occupied by the wax pattern and
6. the investment is broken away (devested).
7. The sprue is removed and the crown is polished and ready for try-in.



Each step in the casting process has an effect on the overall final dimensions of the casting: most castings are slightly oversized to allow for errors in the casting surface and space for cement.



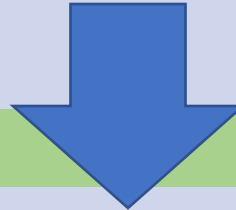
- Dental applications and foundations 11th ed.



If done correctly, casting is accurate, each step in the process influences the final size of the casting (see previous figure).



The accuracy of a cast restoration can be as good as 0.05%. This accuracy translates into a tolerance of less than 20 μm for most dental restorations.

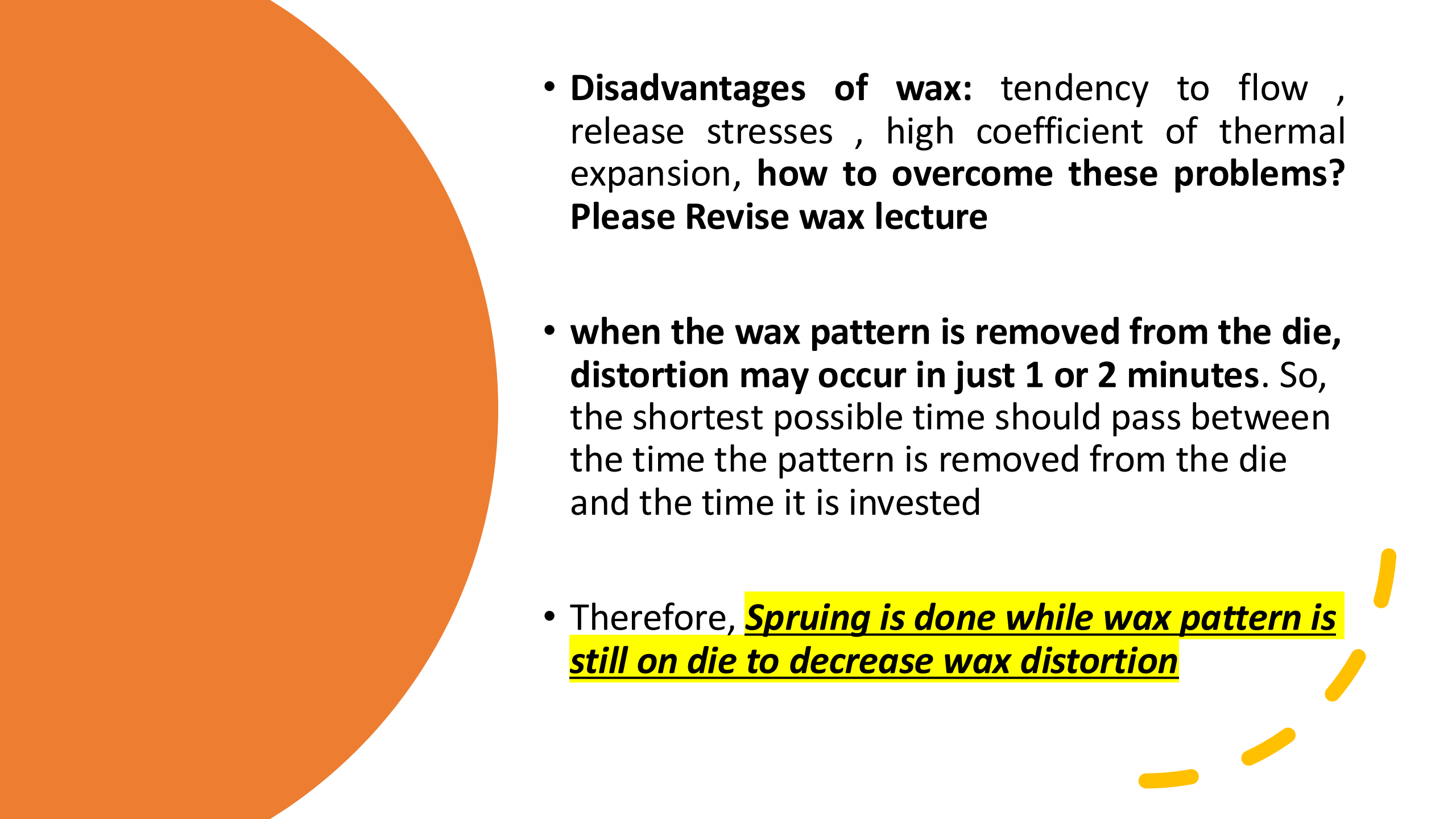


To achieve this degree of accuracy, each step in the casting process must be done carefully

WAXING AND SPRUIING

• Waxing:

- **A wax pattern** is a detailed model of the final dental restoration, including anatomy, contours, occlusal morphology, and proximal contacts
- **Wax** :is easy to manipulate, inexpensive, and easy to be completely eliminated (via burnout) after investing.
- Wax pattern is formed on a **die (model of tooth)(indirect technique of waxing)**
- If wax pattern is formed **on natural tooth itself, then it is called direct technique of waxing**
- Direct technique is rarely used today

- 
- **Disadvantages of wax:** tendency to flow , release stresses , high coefficient of thermal expansion, **how to overcome these problems?**
Please Revise wax lecture
 - **when the wax pattern is removed from the die, distortion may occur in just 1 or 2 minutes.** So, the shortest possible time should pass between the time the pattern is removed from the die and the time it is invested
 - Therefore, **Spruing is done while wax pattern is still on die to decrease wax distortion**



- FIG 12-3 Wax patterns. Wax replicas of teeth to be restored with castings are fabricated on stone dies (A), dies fabricated from impressions of the prepared tooth.

- The dies are removable (B) so that all aspects of the patterns can be inspected.

- When a ceramic–alloy restoration is constructed, the wax pattern is not to full contour but is “cut back” to a thickness that will support the ceramic veneer (C). In this case, the casting is referred to as alloy substructure.

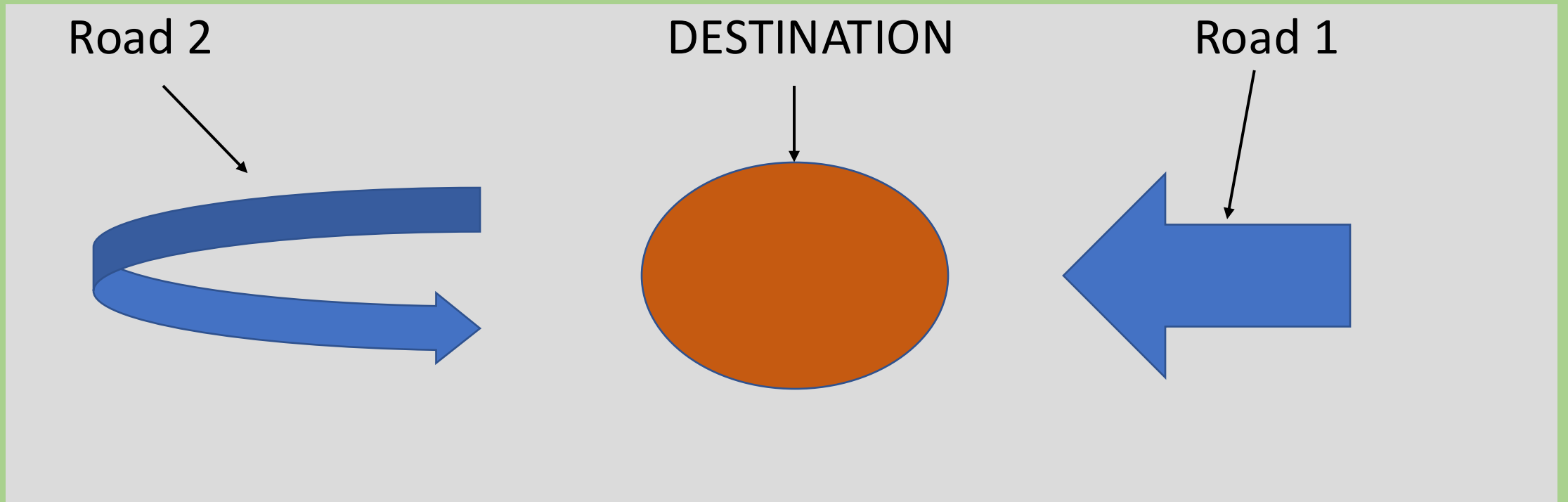
•Spruing

- A sprue forms a channel in the investment through which the molten alloy or ceramic passes to form the restoration.
- Sprues often are wax but also may be plastic or metal.
- If metal, the sprue must be removed before casting because **the metal will not burn out**.
- The decision of **where to place one or multiple sprues** on the wax pattern is complicated and **depends on the type of restoration, the type of casting material, and the experience of the dentist and technician**.
- In the simplest case, a **single sprue** is placed on **one of the cusps** of the wax pattern. In more complex cases, **multiple sprues** may be placed.



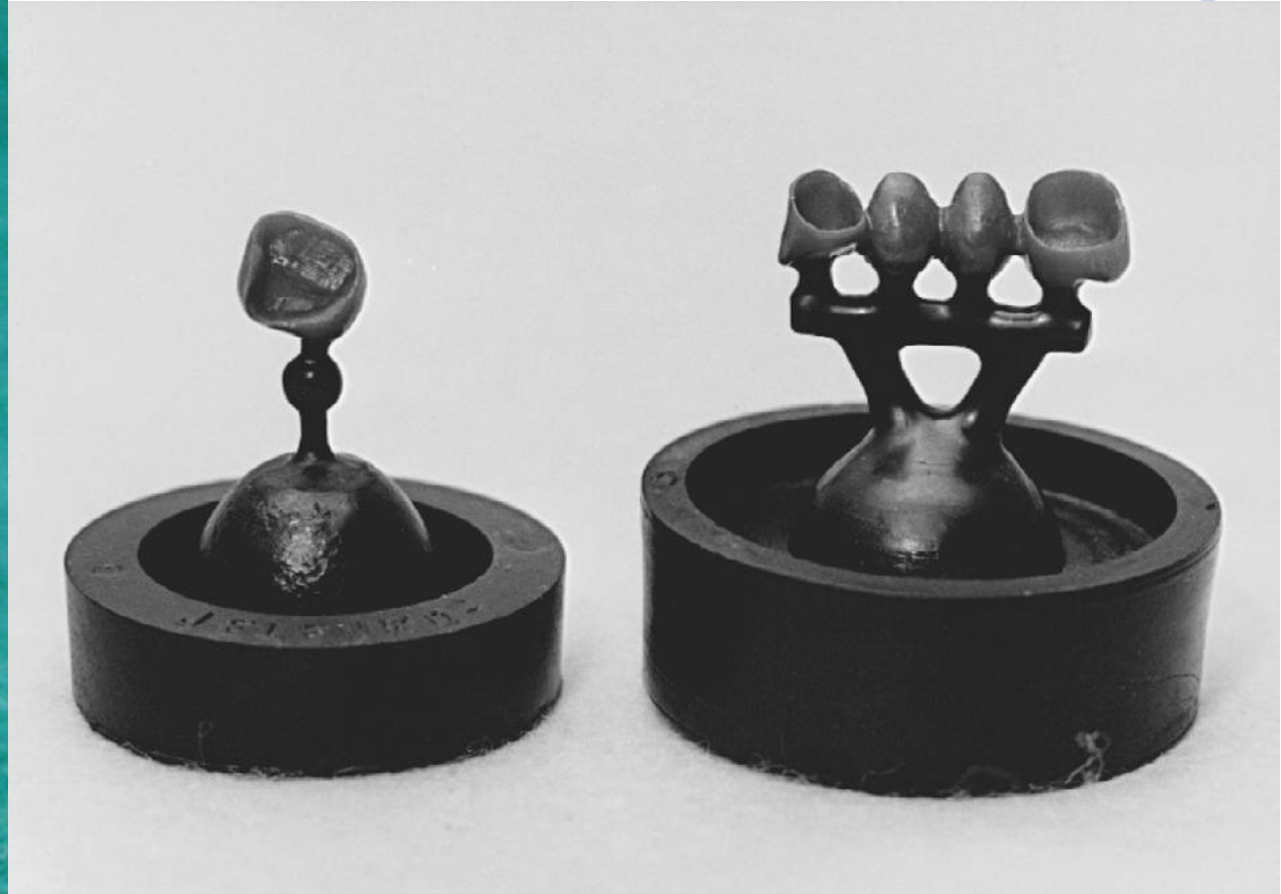
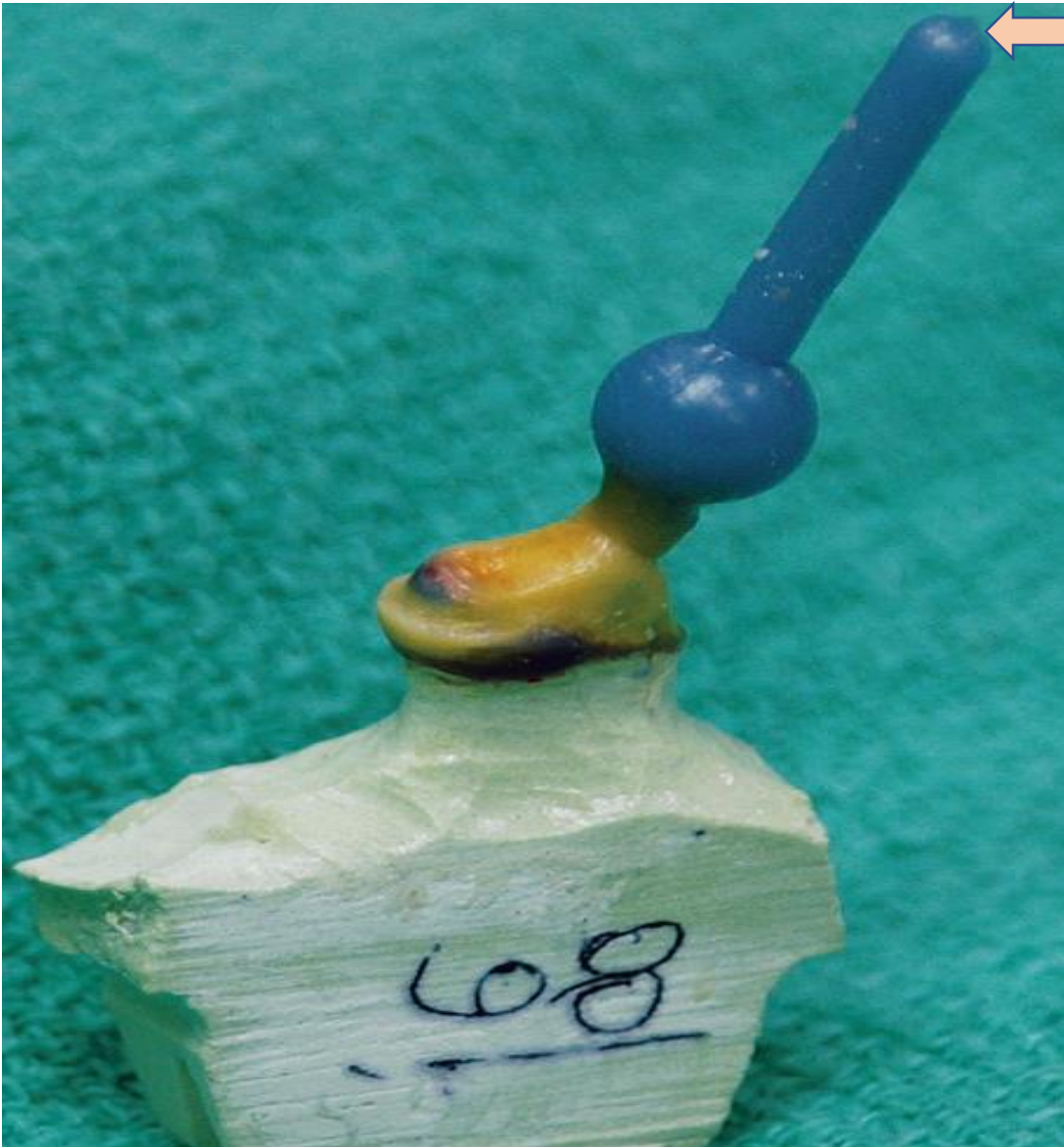
- The other end of the sprue is attached to a **base made of rubber (sprue base)** (Figures 12-1 and 12-4). The sprue is attached at the apex of the cone of the sprue base.
 - The cone would form a depression in the investment to guide the molten alloy into the sprue hole.
- Sprue **diameter and length** vary depending on the type of restoration but should be selected to ensure that the cast material solidifies last in the sprue, and by this strategy the sprue continues to “feed” the casting as it solidifies and shrinks.

To prevent premature freezing in the sprue channel, sprues are generally as big in diameter and as short as possible; a reservoir is often added to the sprue to increase the sprue mass.



Which road will you prefer??????????????

Imagine :The future
road for molten alloy



- Dental applications and foundations 11th ed.

Phillip's science of dental
materials ed. 13 ch.14

Practical

- A reservoir should be added to a sprue network to prevent localized shrinkage porosity (Figure 14-3).
- When the molten alloy fills the heated casting ring during casting , the pattern mold area should solidify first, and the reservoir last.
- The molten metal in the reservoir remains molten to supply additional liquid alloy into the mold as it solidifies.

Reservoir

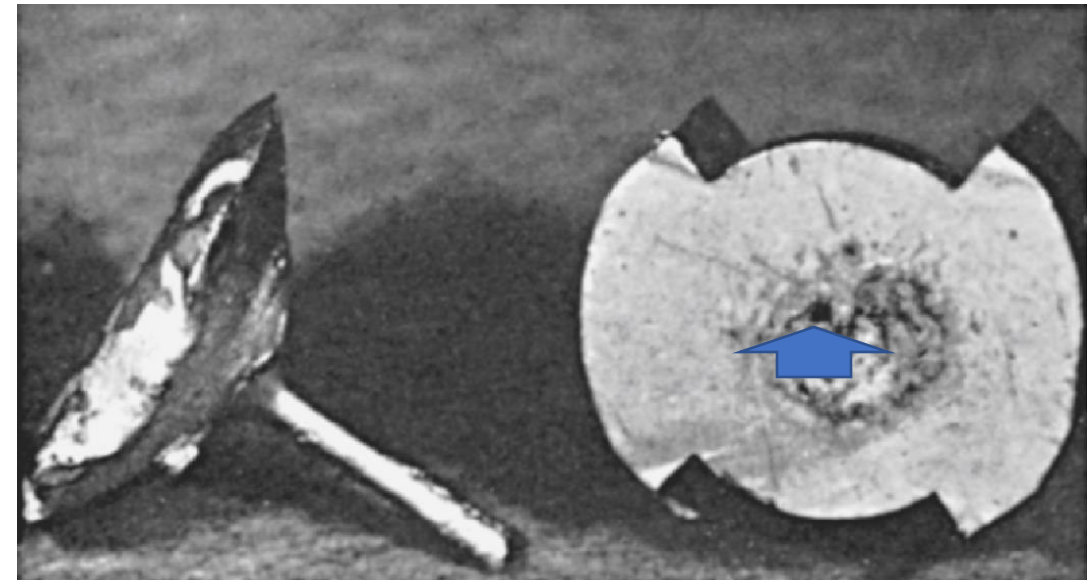
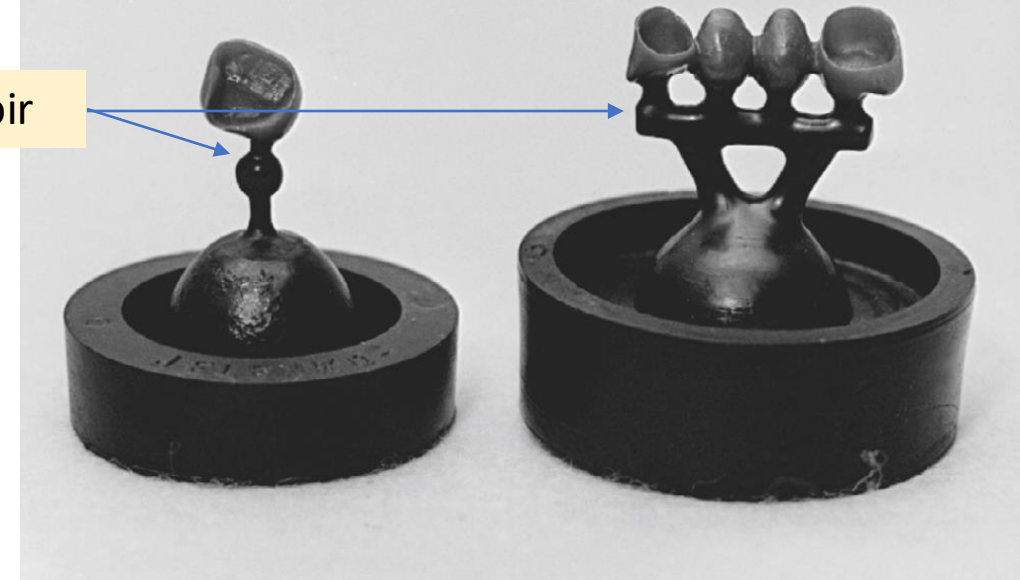
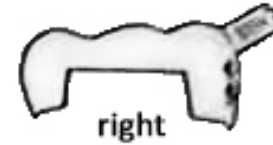


Figure 14-3 Localized shrinkage (*pointer*) caused by using a sprue diameter thinner than required.

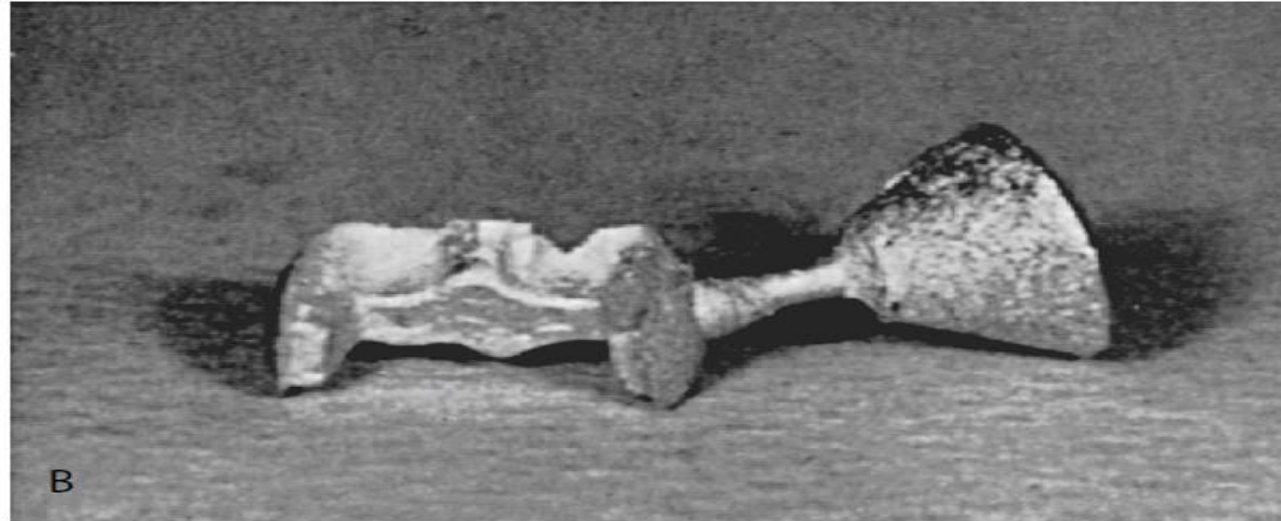
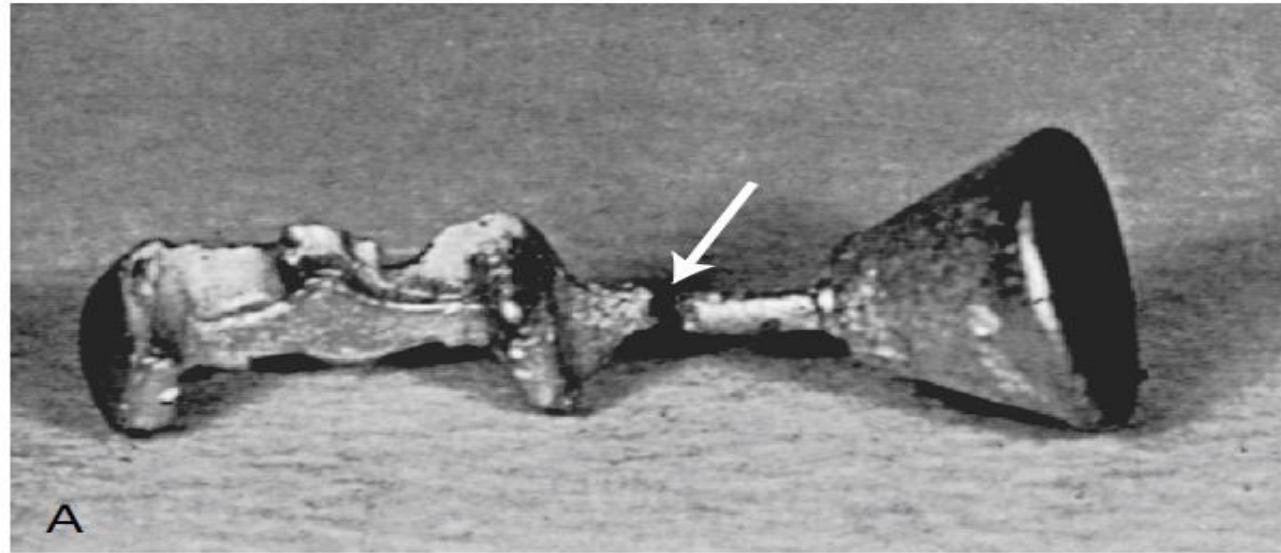
- Sprue is attached to the thickest part of wax with angle 45° to the proximal surface, why? to avoid turbulence during casting, and avoiding casting porosity also.



<https://mydentaltechnologynotes.wordpress.com/2018/08/12/casting-procedures-in-dentistry/>

- sprue thickness must be equal or slightly thicker than thickest part of wax pattern, why? To improve molten alloy flow to mold and provides reservoir during solidification.
- *If sprue is too thin, then it will not act as a reservoir during solidification, and shrinkage spot porosity occurs.*

Practical

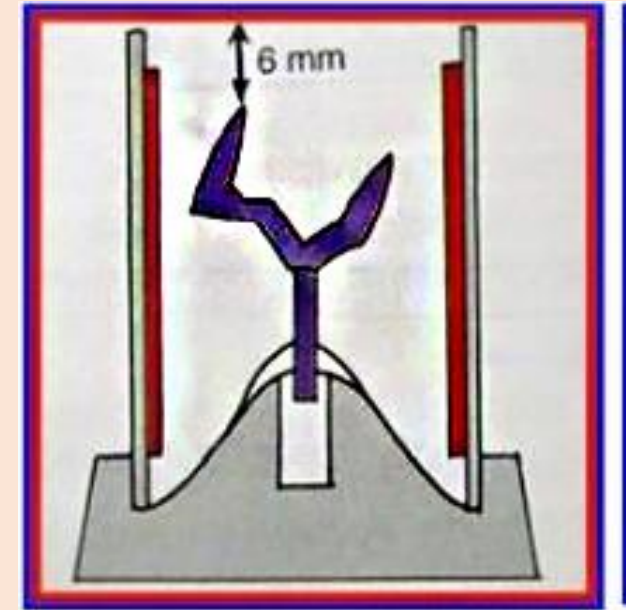


• **Figure 14-4** Effect of sprue orientation. **A**, Detached sprue indicates severe porosity at the point of attachment because of turbulence caused by an improper sprue angle. **B**, Sound casting results with the sprue at approximately 45 degrees to the proximal wall.

Phillip's science of dental materials ed. 13 ch.14

- **Sprue length** : The sprue length should be adjusted so that the top of the wax pattern is within 6 mm of the open end of the ring for gypsum-bonded investments and within 3 to 4 mm of the top of the phosphate-based investment.

- **If sprue too short** : **back pressure porosity** occurs because distance between wax pattern and end of casting ring will be high → air will not escape from mold during casting causing back pressure during molten alloy flow.



<https://mydentaltechnologynotes.wordpress.com/2018/08/12/casting-procedures-in-dentistry/>

- **If sprue too long** : would cause too thin investment between wax pattern and end of casting ring , and Investment may crack during casting causing **fin formation on casting**
Also, very long sprue may cause sprue to solidify first leading to **casting shrinkage and incomplete casting**.

INVESTING AND BURNOUT

- Investments:
- **gypsum bonded Investments** that use gypsum as a binder are used to cast alloys for **inlays, onlays, full-alloy crowns, and some Ni based partial denture frameworks.**
- Gypsum alone (without silica) cannot be used for investing because it lacks sufficient resistance to heat.
- In gypsum bonded Investments the melting temperature of an alloy cannot exceed **1200° C**, and the burnout temperature cannot exceed **700° C** or the gypsum will decompose.
- *phosphate-bonded investments are used for higher melting alloys .*

- **Phosphate bonded investments** are much stronger than gypsum investments and are used for ceramic–alloy restorations, all-ceramic restorations, implant substructures, and partial denture frameworks in which casting alloys have high melting ranges.



• Investing:

- Investing the wax pattern captures all details of the wax pattern into the investment.
- To retain the investment during setting, a casting ring is placed onto the sprue base (Figure 12-4). The casting ring also supports the investment during the casting.
- The ring is lined with a ceramic or paper liner to help the investment to expand during setting and burnout.
- **The liner is moistened before the investment is added to:-)** prevent it from absorbing water from the investment mix .-) Also, that helps in hygroscopic expansion of investment.

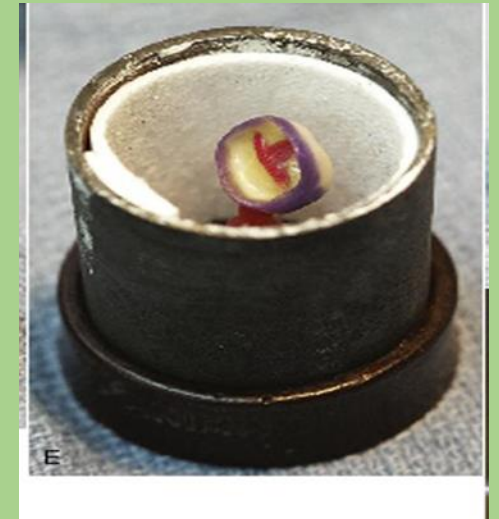
To allow **thermal and hygroscopic** expansion of investment material **a split ring or flexible rubber ring** that permits the setting expansion of the investment can be used



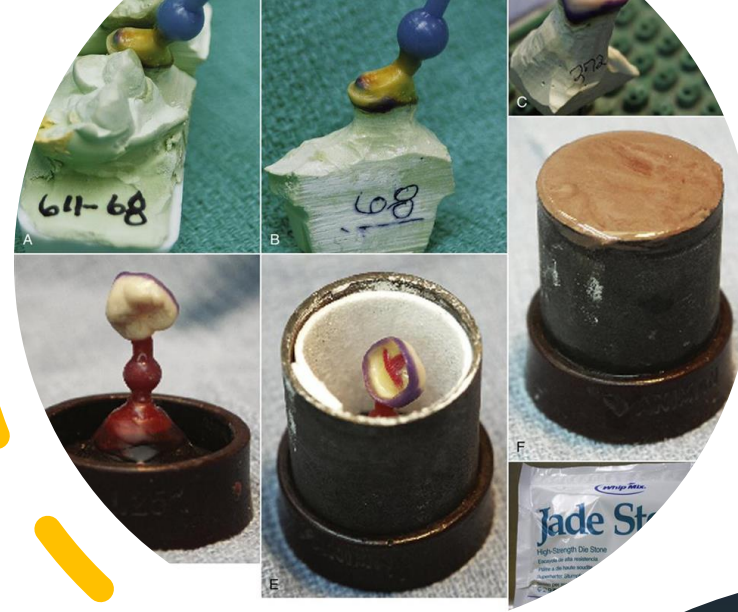
However, the most commonly used technique to provide investment expansion is to line the walls of the **unsplit casting ring with a liner.**

Traditionally, asbestos was the material of choice, **but it can no longer be used because its carcinogenic potential makes it a biohazard.**

Two types of non-asbestos ring liner materials have been produced: **an aluminosilicate ceramic liner and a cellulose (paper) liner.**



- Before investing, the wax pattern is treated with a **surfactant** that aids flow of the water-based (hydrophilic) investment material over the water-hating (hydrophobic) wax pattern.





- Once mixed, the investment is **poured slowly** into the casting ring, taking extreme care to flow the investment around the wax pattern without forming bubbles or voids (**vibrator is used**)
- Pouring till ring is **full**
- **Vacuum mixer of investment can be used for vacuum investing**
- **Vacuum investing is better than hand investing.**
- Because the wax pattern is dimensionally unstable once removed from the die, the time taken to invest should **be less than 10 minutes**.
- The invested wax pattern is allowed to set for **at least 45 minutes** before burnout but **should not be allowed to desiccate**.

<https://pocketdentistry.com/12-casting-and-soldering/>

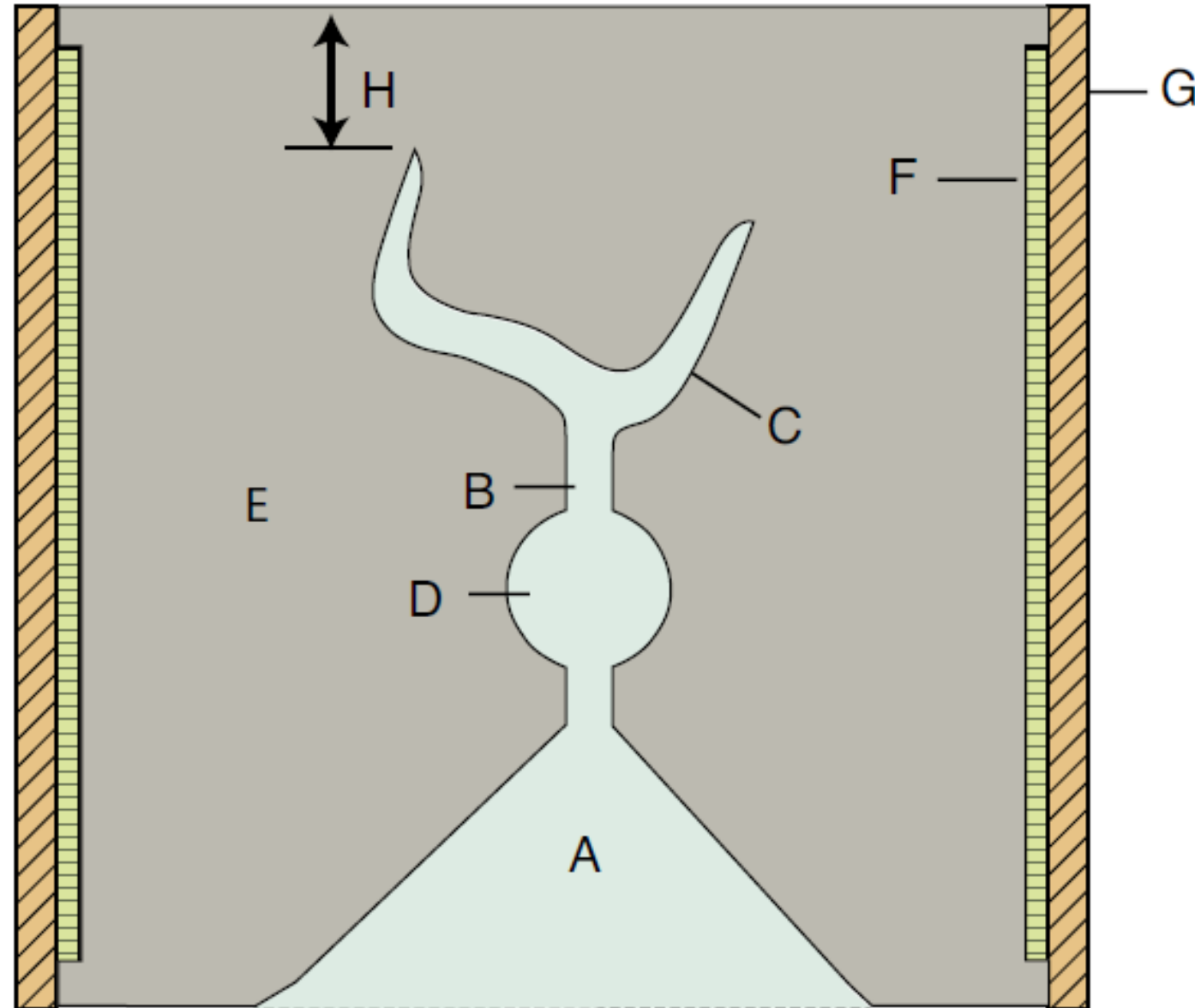
Summary



FIG 12-4 Sequence of steps for investing. Whether the wax pattern is a cut-back (A and B) or to full contour (C),

Practical

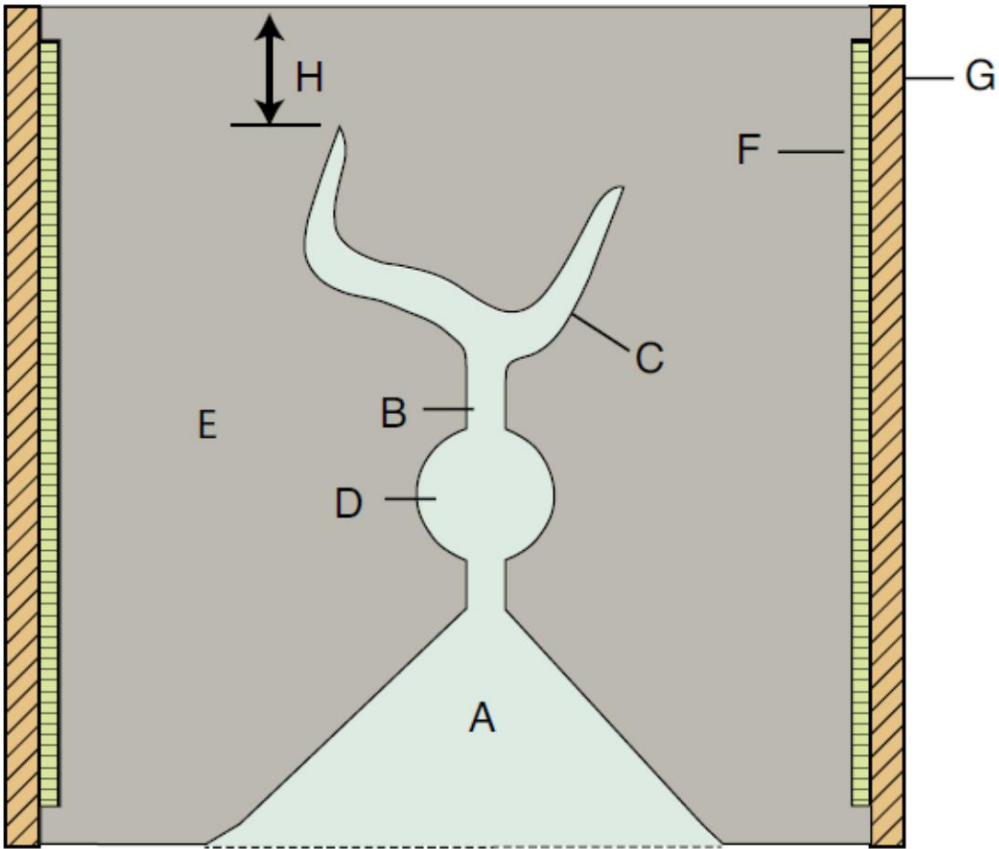
Write correct labels



Practical

- A, Crucible former. B, Sprue. C, Cavity formed by wax pattern after burnout.

D, Reservoir. E, Investment. F, Liner. G, Casting ring. H, Recommended maximum investment thickness of approximately-----
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Burnout

- Once the investment has set (after approximately 45minutes), the casting ring is placed into an oven to burn out the wax pattern and sprue
- wax does not melt but combusts to allow complete removal.
- After burnout, a space remains that was occupied by the wax pattern and sprue.
- So,The sprue space forms a channel through which the molten alloy can flow to the space formerly occupied by the wax pattern



Burnout oven. The invested wax pattern (without the sprue base) is placed into an oven

(A) to burn out the wax prior to casting. The oven maintains a temperature compatible with the investment used .

(Dental materials foundations and applications 11 ed)

- Burnout also cause thermal expansion of the investment.
- burnout provides temperature compatibility between the hot, molten alloy and the investment, **why?:-** If the molten casting material were cast into cold investment at room temperature, the rapid change in temperature can cause cracking to the investment.
 - The relatively cold investment also would cause the casting material to solidify prematurely (early) before the restoration space was completely filled.
- A rule of thumb is that the melting temperature (liquidus) of an alloy should not be more than 500° C higher than the burnout temperature.

- Once the casting ring and investment are heated, the ring should be cast immediately before it cools to any degree; any delay of more than 20 to 30 seconds will compromise the fit and quality of the cast restoration.

- The thermal expansion of the investment is not reversible, and if the ring cools before it is cast, either the investment will crack or the casting will be distorted.

Riyadh Colleges Of Dentistry and Pharmacy

Department of Prosthodontics



Educational Video Series

References

- 1- Dental applications and foundations 11th ed.
- 2-
<https://mydentaltechnologynotes.wordpress.com/2018/08/12/casting-procedures-in-dentistry/>
- 3- Philips 's science of dental materials 13ed. CH.14

The background is a complex, abstract geometric pattern. It consists of numerous 3D cubes and 2D hexagons arranged in a way that creates a sense of depth and movement. The colors used are a variety of shades: deep teal, vibrant purple, bright blue, and light green. Some shapes are solid, while others are semi-transparent, allowing the underlying pattern to show through. The overall effect is a modern, digital aesthetic.

Thank you